A Gauge Theory of Line Defects in an Elastic Continuum: Dislocation Rings, the Peach–Koehler Force, and Emergent Interaction Laws

(A Geometrodynamic Model of Particles as Elastic Defects)

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Abstract

We present a consistent account of a geometrodynamic model in which elementary particles are treated as topological defects (dislocation and disclination rings) of an elastic aether, and the fundamental interactions appear as different projections of a single elastic response of the medium. The model is built on the Kadić–Edelen and Lazar gauge theory of dislocations [2, 3], in which the magnetic field is identified with the dislocation density and the Peach–Koehler force plays the role of the Lorentz force. Gravity is introduced through a "Fermi film" mechanism — a deformable boundary of the action integral held orthogonal to the curved world tubes of accelerating charges (a generalization of Fermi's 1923 treatment of the 4/3 paradox [1]). We show that the inverse-square law for gravity is a *theorem* (a consequence of the Laplacian Green function), not a fit. The model is formulated in an elastic unit basis (L, T, M), without invoking electric charge as a primary entity. The limits are drawn honestly: the *structure* of the interactions (power laws, signs, hierarchy) is derived, together with a set of relations between scales, whereas the absolute values of the fundamental constants (α, G) remain input parameters. The paper is a working preprint intended as a first introduction to the subject.

Keywords: gauge theory of dislocations, aether, Fermi film, magnetic monopole, Burgers vector, Peach–Koehler force, 4/3 paradox, geometrodynamics.

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1 Introduction

1.1 Motivation: the particle as a defect of the medium

The idea that elementary particles may be not point objects but extended structures in some medium has a long history — from Kelvin's vortices to modern soliton models. The present work develops a concrete version of this programme: *a particle is a topological defect of an elastic continuum* (the "aether"), namely a closed dislocation line (ring) or its generalizations. The approach is attractive because it rests on the well-developed mathematical apparatus of the theory of crystal defects [9, 3] and its gauge formulation [2].

1.2 Why dislocations

In crystals, dislocations are line defects characterized by the Burgers vector $\mathbf{b}$, which measures the closure failure of a circuit around the defect line. M. Lazar [2] showed that the dynamic gauge theory of the translation group $T(3)$ leads to equations structurally identical to Maxwell's equations (Table 1). This provides a physically meaningful "dictionary" between defect theory and electrodynamics.

Table 1: Correspondence Maxwell $\leftrightarrow$ dislocation theory (after Lazar [2]).

Electrodynamics	Dislocation theory
Magnetic field $\mathbf{B}$	dislocation density T^a
Electric field $\mathbf{E}$	dislocation current I^a
Magnetic excitation $\mathbf{H}$	pseudomoment stress H_a
Magnetic flux $\oint \mathbf{B}\, d\mathbf{S}$	Burgers vector $\oint T^a = b$
Lorentz force	Peach–Koehler force

1.3 The key feature: the charge is magnetic, not electric

In the proposed model *electric charge is not a primary entity.* What is primary is the magnetic charge, identified with the flux of the dislocation density, i.e. with the Burgers vector. The electron is built as a precessing magnetic dipole. This is consistent with the long-standing line connecting dislocations and Dirac monopoles [10].

1.4 Structure of the paper

Section 2 introduces the elastic unit basis. Section 3 describes the particle as a precessing dipole. Section 4 presents the Fermi-film mechanism and the derivation of gravity. Section 5 collects the six interaction channels. Section 6 honestly delimits the model.

2 The elastic unit basis

We build a system of units from three primitives of the elastic medium: length L, time T and mass M. There is *no* electric charge and no ampere in the basis. The main quantities are:

- Burgers vector b (lattice displacement) — dimension $[L]$;
- shear modulus μ — dimension $[ML^{-1}T^{-2}]$ (Pa);
- density ρ — $[ML^{-3}]$;
- speed of sound/light $c = \sqrt{\mu/\rho}$ — $[LT^{-1}]$.

The magnetic charge as the flux of the dislocation density $q_m = \oint T^a\, dS$ has the dimension of length $[L]$ — this is internally consistent in the elastic theory (unlike in SI, where the magnetic charge is measured in A·m; transferring SI units to the elastic medium is unnecessary and leads to confusion).

3 The particle as a precessing dipole

3.1 Geometry

The electron is modelled as a pair of dislocation rings (of radius a) whose planes are *fixed*, while along each ring a "special point" — a localized magnetic charge $q_m = \Delta b$ (a jump of the Burgers vector) — runs with frequency ω. By "precession" we mean precisely the motion of the special point along the fixed ring, not a wobbling of the planes. The two rings are separated by a distance d and carry opposite charges $\pm q_m$ (the N and S poles), forming a magnetic dipole.

3.2 Mass from self-interaction

Mass is not introduced separately but is the elastic self-interaction energy of the defect:

$$m = \frac{\mu\, b^2 a}{c^2}, \tag{1}$$

which has the correct dimension $[M]$.

3.3 Magnetic moment and spin

The magnetic moment is a *static* dipole moment $\mu_{\mathrm{mag}} = q_m\, d$ (not a current-loop moment: the ring current is magnetic, and the naive current-loop formula does not apply here). Spin is tied to topology: a pair of ordinary rings has configuration period 2π (a boson), whereas a pair of Möbius rings (ring + anti-ring) has period 4π, corresponding to a fermion of spin 1/2. In the Möbius pair the two rings carry *opposite* directions of the Burgers vector ($+\Delta b$ and $-\Delta b$, the N and S poles) — hence a static magnetic dipole. Crucially, the Burgers twist $b(\varphi) = \cos(\varphi/2)$ is a Frank rotation by π per circuit, i.e. a *disclination* ($SO(3)$) degree of freedom: the Möbius sector of the model already lies in $SO(3)$, and the magnetic moment $\mu = \nu = n/2$ follows as the flux of the spin connection (Osipov [6]). Hydrodynamically the same object is a vortex: the vorticity corresponds to the Frank density, and the precession ($\omega > 0$) required for stability is the circulation of that vortex.

3.4 Stability and annihilation

Opposite charges $\pm q_m$ attract and, without motion, would collapse. Precession provides a centrifugal barrier (conservation of angular momentum $J = ma^2\omega$) that stabilizes the pair at a finite distance:

$$E(d) = -\frac{A}{d} + \frac{J^2}{2md^2}, \qquad d_{\mathrm{eq}} = \frac{J^2}{mA}. \tag{2}$$

As $\omega \to 0$ the barrier vanishes — the pair annihilates (Frenkel recombination, the total Burgers vector going to zero).

4 The Fermi film and the derivation of gravity

4.1 Historical context: the $4/3$ paradox

E. Fermi in 1923 [1] showed that the well-known discrepancy between the electrodynamic ($\frac{4}{3}U/c^2$) and relativistic (U/c^2) masses of a charged body is removed if the boundary of the action integral is taken *strictly orthogonal* to the curved world tube of the accelerating charge. We call this deformable boundary the *Fermi film*; formally it is analogous to the Gibbons–Hawking–York boundary term in general relativity [8].

4.2 The source of deformation

An accelerating charge curves its world tube; the film, remaining orthogonal, is deformed. The density of the deformation source is proportional to the *square* of the proper acceleration $W = \omega^2 a$:

$$\sigma = \mu\, b^2\, \frac{\langle W^2 \rangle}{c^2} > 0. \tag{3}$$

The sign is always positive (a square), so gravity is always attractive and does not depend on the sign of the charge (universality).

4.3 The inverse-square law as a theorem

Theorem 1 ($1/R^2$ law)**.** *If the film deformation Δ obeys the equilibrium equation of an elastic membrane with a point source,*

$$\gamma\, \nabla^2 \Delta(\mathbf{r}) = -\sigma\, \delta^3(\mathbf{r} - \mathbf{r}_0), \tag{4}$$

then the solution is the Laplacian Green function, $\Delta(R) = \sigma/(4\pi\gamma R)$, whence the interaction potential $U(R) \propto 1/R$ and the force $F(R) \propto 1/R^2$.

The exponent follows from the three-dimensionality of space, not from a fit.

4.4 Stiffness parameters: two scales

What μ_{aeth} is. The quantity μ_{aeth} is the *shear modulus of the vacuum*, the stiffness of the aether with respect to shear deformation, of pressure dimension [Pa]. It is through this elastic modulus that, in the (L, T, M) basis (without the ampere), the main quantities of the model are defined:

$$c = \sqrt{\mu_{\text{aeth}}/\rho} \qquad \text{(transverse-wave speed)}, \tag{5}$$

$$m = \mu_{\text{aeth}}\, b^2 a/c^2 \qquad \text{(particle mass)}, \tag{6}$$

$$F = \mu_{\text{aeth}}\, b^2/R \cdot (\text{geometry}) \qquad \text{(Coulomb force)}. \tag{7}$$

Physically μ_{aeth} governs shear deformations (dislocations), from which charge and mass are built; hence it sets the physics at the particle scale (of order the Compton length λ_C). Its numerical value is obtained by *calibration*: substituting the electron mass m_e and its

classical radius r_e into the self-energy of the dislocation ring and solving for $\mu_{\rm aeth}$. This is *not* a derivation from first principles but the fixing of a single number by the electron, which we state explicitly.

A check against the model shows that *one* modulus cannot account for both mass/Coulomb and gravity: the required values differ by ∼84 orders of magnitude. Hence two scales are needed:

$$\mu_{\rm aeth} \approx 2.7 \times 10^{28}\ \text{Pa} \qquad \text{(EM/mass, Compton scale)}, \tag{8}$$

$$\gamma_P = \frac{c^7}{8\pi\hbar G^2} \approx 1.8 \times 10^{112}\ \text{Pa} \qquad \text{(gravity, Planck scale)}. \tag{9}$$

The gravitational constant G enters γ_P: it is an *input* parameter of the aether stiffness, not derived from more fundamental quantities.

What γ_P is, and the nature of the hierarchy. The Planck modulus is the Planck *energy density* divided by 8π:

$$\gamma_P = \frac{c^7}{8\pi\hbar G^2} = \frac{1}{8\pi}\,\frac{m_P c^2}{\ell_P^3}, \qquad \ell_P = \sqrt{\hbar G/c^3}, \quad m_P = \sqrt{\hbar c/G} \tag{10}$$

(numerical agreement $< 1\%$). This is not an arbitrary combination but the stiffness of the vacuum at the Planck scale; the factor 8π is the same as in the Einstein equations. Likewise $\mu_{\rm aeth} = m_e c^2/(r_e^2 \lambda_C)$ is the energy density at the electron scale. The ratio of the two moduli

$$\frac{\gamma_P}{\mu_{\rm aeth}} \sim \frac{m_P}{m_e}\ \text{(with a volume factor)} \sim 10^{84} \tag{11}$$

reduces to the ratio of the Planck and Compton scales, i.e. to the smallness of m_e/m_P. Thus the hierarchy of the two stiffnesses is *equivalent* to the standard *hierarchy problem* ($m_e \ll m_P$): the model does not solve it, but neither does it introduce a new arbitrary parameter — it inherits the well-known smallness of m_e/m_P.

5 The six interaction channels

All interactions are projections of a single elastic response. Their decay laws and sources are collected in Table 2; the power laws are structural consequences (Green function, dipole–dipole interaction, mediator mass, confinement), not the result of a fit.

Table 2: The six interaction channels of the model.

Channel	Law	Source	Coupling
Coulomb	$1/R^2$	product of Burgers charges	$\alpha = 7.30 \times 10^{-3}$
Magnetic	$1/R^4$	dipole–dipole $q_m d$	$\sim \alpha/4$
Gravity	$1/R^2$	$\langle W^2 \rangle$ of the Fermi film	$\alpha_G = 1.75 \times 10^{-45}$
Exchange	$1/R^3$	precession phase $\cos 2\Delta\phi$	$\sim \alpha$
Weak	e^{-R/R_W}	twist chirality	$\sim 1/30$
Strong	$\propto R$	fractional $b/3$ + ribbon	~ 1

The whole system closes on two dimensionless numbers:

$$\alpha = \frac{ke^2}{\hbar c} = \frac{1}{137.04}, \qquad \alpha_G = \left(\frac{m_e}{m_{\rm Planck}}\right)^2 = 1.75 \times 10^{-45}, \tag{12}$$

whence the weakness of gravity relative to the Coulomb force: $\alpha_G/\alpha \approx 2.4 \times 10^{-43}$.

6 Limits of the model: what is proved, assumed, open

In the spirit of scientific honesty we explicitly separate the levels of certainty.

Proved (structurally). The decay laws of all channels; the attractiveness and universality of gravity; the dimensional closure of mass; the consistency of the magnetic charge as a length-dimension quantity in the elastic basis; the annihilation mechanism and the protection by precession; the Frank vector $\Omega = \pi$ of the $\cos(\varphi/2)$ twist (a wedge half-disclination, $\nu = 1/2$, spin 1/2).

Assumed (input parameters). The constants α and α_G; the gravitational constant G (embedded in the stiffness γ_P); the existence of two stiffness scales and their ratio.

Open. The origin of the numerical values of α and α_G; a rigorous derivation of the gyromagnetic ratio $g = 2$ (in the classical ring picture one gets $g = 1$; $g = 2$ requires a spinor level of description in the spirit of [6]); the dimensional factor in the gravity source, which requires a dynamical (time-dependent) film stiffness; the numerical values of the weak- and strong-sector couplings. The tensor ($J = 2$) channel of the trace-free quadrupole Q_{ij} exists structurally (spin-2) but gives only a short-range $1/R^5$ correction; the long-range $1/R^2$ gravity is intrinsically scalar (the trace $\langle W^2 \rangle$).

7 Conclusion

We have proposed an internally consistent model in which particles are elastic defects of the aether, and the interactions are projections of a single medium response through the Peach–Koehler force and the deformation of the Fermi film. The model explains the *structure* (laws, signs, hierarchy of interactions) without claiming to derive the fundamental constants from first principles. Such a formulation makes the theory falsifiable: for magneto-gravitational devices, for example, it predicts a negligible ($\sim 10^{-13}$) relative correction to weight, incompatible with claims of macroscopic antigravity. Further work includes a rigorous spinor derivation of $g = 2$, a refinement of the dynamical film stiffness, and this English version for archival publication.

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